TEACHER NOTES · 2 WEEKS

Can a computer spot a fake planet?

Teacher's guide to the Kepler false-positive project

> **Students measure what fraction of Kepler's false positives can be identified from the light curve alone — and prove that the rest cannot be.**

The project is deliberately built so that the final answer is a limit rather than a score. Students train a Random Forest on ten features they measured themselves, then show that the objects it misses are concentrated in the one category of false positive that photometry cannot detect, because the light curve records how much light was lost but not where. This gives a genuine result that does not depend on how good their model turned out to be, so weaker groups still reach a defensible conclusion.

LEARNING OBJECTIVES

Scientific	Students measure a real limit of a real dataset: the fraction of Kepler false positives that photometry alone can identify, and the physical reason for the remainder.
Statistical	Students learn why accuracy is the wrong metric for unbalanced classes, and use precision, recall and a precision-recall curve instead.
Methodological	Students discover data leakage themselves rather than being warned about it, and learn to group a train/test split by object rather than by row.
Practical	Students build a working measurement pipeline: download, clean, detrend, fold, measure, and validate the measurement on injected signals.



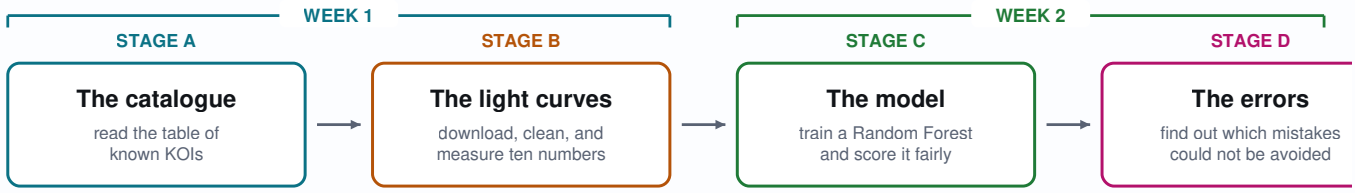
Two stars eclipsing each other nearby produce the same dip in your light curve. No feature measured from brightness alone can tell them apart.

The single idea to protect. If students take away only one thing, it should be that a very high score is a reason to look for a mistake. Stage A is designed to produce a suspiciously good result on purpose, so that the lesson arrives as a discovery rather than as a warning.

TIMING

How the two weeks fit together

Four taught sessions of about an hour, plus one unsupervised download.



Each stage matches one Part of the notebook.

Stage D is where you find the scientific result. Stages A to C build the tools you need to get there.

WEEK	SESSION	WHAT HAPPENS
Week 1	**Session 1** 60–75 min	**Stage A, notebook Part 1** Download the catalogue, plot the class distributions, run the leakage audit, train the catalogue baseline, then break it deliberately with `koi_score`.
	Session 2 60–75 min	**Stage B, first half** Download and clean one light curve together as a class. Write the first four feature functions and test them on the synthetic light curve provided in the notebook.
	Homework unsupervised	**Stage B, download loop** Start the measurement loop running. It checkpoints after every ten targets, so it can be stopped and resumed. Expect 20–40 minutes for 100 targets.
Week 2	**Session 3** 60–75 min	**Stage B remainder + Stage C** Complete the three TODO features, then train the Random Forest, read the classification report and draw the precision-recall curve.
	Session 4 60–75 min	**Stage D, notebook Part 4** Attach the false-positive flags, compare caught against missed objects, and draft the conclusion paragraph.
	Optional 30 min	**Presentation** Each group presents one folded light curve their model got wrong, and explains why.

The one scheduling risk

Stage B depends on downloading about a hundred light curves, which takes 20 to 40 minutes and several gigabytes. If that cannot happen between Session 2 and Session 3, the project stalls. The safe approach is to run the download yourself in advance and keep `measured_features.csv` in reserve. Students who have their own measurements use those; anyone whose download failed uses yours and still completes Stages C and D.

If you only have one week

Distribute a pre-measured `measured_features.csv`, skip the download loop entirely, and run Stage A, Stage C and Stage D. Students lose the measurement practice, which is a real loss, but the scientific conclusion in Stage D survives intact.

DELIVERABLES

What to collect at each stage

These are deliberately absent from the student handout, so that you can set and weight them yourself.

STAGE A — The catalogue, and the clue hidden in it

> **Deliverable.** A short written note: the accuracy of the second model, the column it relied on, the list of columns you set aside with a reason for each, and why a very high score can be a warning sign rather than a good result.

STAGE B — From brightness to numbers

> **Deliverable.** The file `measured_features.csv` with at least 80 objects measured; four folded plots with the windows drawn on and one sentence under each saying what gives that object away; and a count of how many targets failed to download and why.

STAGE C — Training the model

> **Deliverable.** The classification report, the confusion matrix and the precision-recall curve, all from your own measured features; plus two or three sentences on whether you beat the Stage A result and whether the most important features are the ones the physics predicts.

STAGE D — Finding the limit

> **Deliverable.** The bar chart comparing the four flags between caught and missed objects, and a written conclusion in the form given in the notebook.

ASSESSMENT RUBRIC

STRAND	WHAT A STRONG SUBMISSION LOOKS LIKE
Method	The split is grouped by star, the metric suits the class imbalance, and the feature functions were tested on an injected signal before being used on real data.
Honesty	Failed downloads, dropped objects and missing values are reported rather than quietly removed. The list of banned columns is written down with reasons.
Physics	The important features are explained in terms of what an eclipsing binary actually does, not simply read off the importance plot.
Conclusion	A number with an uncertainty, a comparison against the guessing baseline, and a stated physical reason for the errors that remain.
Presentation	Plots have axis labels and units. The folded light curves shown are the ones that support the argument being made.

> **Weighting suggestion.** Put most of the credit on Method, Honesty and Conclusion rather than on the recall the model achieved. A group that reports 0.62 recall with a correct grouped split and an honest error analysis has done better work than a group reporting 0.98 with a leaked column.

THE SIX TODOS · PART 1

The three feature functions

These are where most of the learning happens. Worked solutions are in the teacher notebook.

TODO 1 — Odd-even depth difference (Stage B)

What it tests. Tests whether students can build and combine boolean masks, and whether they understand that an eclipsing binary alternates deep and shallow eclipses.

Expected answer. Measure the depth using `core & (ep % 2 == 1)`, then again with `== 0`, and divide the difference by `std(flux[out]) / sqrt(N_in / 2)`.

Common errors. The most common error is dividing by the raw scatter rather than the standard error, which makes every object look consistent. A second common error is using `full` instead of `core`, which dilutes the difference.

TODO 2 — Secondary eclipse depth (Stage B)

What it tests. Tests understanding of what folded phase means, and that the folded range runs from –P/2 to +P/2 so half a period away appears at *both* edges.

Expected answer. Build the mask `abs(abs(ph) - period/2) < 0.5 * duration` and pass it to `_depth()`. Return 0.0 if fewer than three points land there.

Common errors. Students often write `abs(ph - period/2)`, which catches only one edge and halves the number of points. Ask them to plot the mask to see the problem.

TODO 3 — Implied stellar density (Stage B)

What it tests. Tests careful unit handling, which is where most of the marks are genuinely earned.

Expected answer. $a/R_* = P/(\pi T)$ with P and T both in seconds, then $\rho_* = 3\pi(a/R_*)^3/(GP^2)$ with $G = 6.674\times10^{-8}$.

Common errors. Almost everyone tries it in days first and gets an answer wrong by about 10^{15}. Tell them to check a Sun-like star comes out near 1.4 g/cm^3 before moving on.

Before students touch real data

The notebook provides `fake_lightcurve()`, which builds a synthetic light curve with a transit of known depth, shape, secondary eclipse and odd-even difference. Insist that students get the correct answers out of the synthetic planet and the synthetic binary before they run anything on a real star. A fake planet should give `v_shape` near 1.07, a secondary near zero, and a small odd-even value. A fake binary should give `v_shape` near 1.4, a clearly positive secondary, and a large odd-even value.

THE SIX TODOS · PART 2

Splitting, plotting and the final analysis

TODO 6 is the one to spend time on. It is where the project becomes science.

TODO 4 — Grouped train/test split (Stage C)

What it tests. Tests whether the leakage lesson from Stage A actually transferred.

Expected answer. `GroupShuffleSplit(n_splits=1, test_size=0.30, random_state=RANDOM_STATE)` then `next(splitter.split(X, y, groups=groups))`, followed by the assertion that the two sets of kepids do not intersect.

Common errors. Some students will use `train_test_split` out of habit. The assertion in the notebook will fail, which is the intended teaching moment.

TODO 5 — Scatter plot of the two strongest features (Stage C)

What it tests. Tests whether students can read a two-class scatter plot and say honestly whether the classes separate.

Expected answer. Group by `is_fp`, scatter `v_shape` against `secondary_ppm.clip(-500, 5000)`, label both axes and add a legend.

Common errors. Watch for students concluding the classes separate cleanly when the clouds clearly overlap. Ask them to estimate what fraction of points sit in the overlap.

TODO 6 — The flag comparison (Stage D)

What it tests. This is the assessment centrepiece. It tests whether students can turn a confusion matrix into a physical statement.

Expected answer. Merge the four flag columns onto the test set on `kepoi_name`, split the true false positives by `pred`, take the mean of each flag in each group, and plot the two columns side by side.

Common errors. If the test set is small there may be too few missed objects to say anything. The notebook prints a warning below three. In that case have them pool the cross-validated predictions instead of using the single split.

DIAGNOSING A STUCK GROUP

WHAT THEY REPORT	WHAT TO CHECK FIRST
Recall above about 0.95	Look for a mistake before celebrating. Check that no catalogue column, flag or name has got into the feature list, and that the split is grouped on `kepid`.
Recall below about 0.5	This is usually a measurement problem rather than a model problem. Plot the folded light curves of a few objects the model got wrong and check that the features were measured on a real transit.
One feature is far more important than all the others	Either you have found something real, since depth and secondary depth do carry a lot of the signal, or that feature contains the answer. Plot it against the label and decide which.
The model gives the same answer for every object	Your sample probably became very unbalanced after removing rows with missing values. Check how many rows survived `dropna`.
Many features come out as NaN	Too few data points fell inside one of the windows. This usually means a long-period object with only two or three transits, or a wrong epoch. Check that `koi_time0bk` and the light curve use the same time system, which is BKJD for both.

PREPARATION

Before the first session

Six things to do in advance, and four questions students reliably ask.

CHECKLIST

- Run the teacher notebook end to end at least once on your own machine, ideally a week ahead. The download step is the only part that can fail unpredictably.
- Check that the NASA Exoplanet Archive and MAST are both reachable from the school network. Some networks block the ports that `astroquery` uses.
- If bandwidth is limited, download one shared sample yourself and distribute `measured_features.csv` so that Stage C and Stage D can go ahead regardless.
- Decide before Session 1 whether students work individually or in pairs. Pairs work well because Stage B involves a lot of waiting.
- Have the answer to `koi_time0bk` ready: it is in BKJD, which is BJD minus 2454833, and it is the same time system lightkurve returns. This causes more confusion than anything else in the project.
- Keep the teacher notebook to yourself until after Session 3. The three feature TODOs are where most of the learning happens.

QUESTIONS STUDENTS ASK

“Why can we not just use the catalogue depth and period?”

You can, and that is exactly Stage A. The point of Stage B is that those numbers were produced by the same pipeline that decided the answer, so using them does not show that the light curve itself carries the information.

“My model is worse than the Stage A baseline. Have I failed?”

No. The Stage A model had the benefit of professionally fitted parameters from the full mission dataset. Matching it with ten hand-measured features on a hundred objects is a good outcome, and explaining the gap is worth more than closing it.

“Why is the centroid flag not just another feature we could measure?”

Because it is not in the data we downloaded. A light curve is a single brightness value per time step; the position information lives in the target pixel files. This is the whole point of Stage D, and the extension task is to go and get those pixels.

“Can we use a neural network instead?”

Not on a hundred objects. A CNN needs tens of thousands of labelled light curves. The artificial transit extension is the route to that, because injected signals give unlimited reliable labels.

FILES `KOI_false_positives_TEACHER.ipynb` is the fully worked notebook. `KOI_false_positives_STUDENT.ipynb` is the version with the six TODOs. `KOI_ML_project_handout.pdf` is the student handout.

Both notebooks are generated from one source file, so the only differences between them are the six TODO sections.